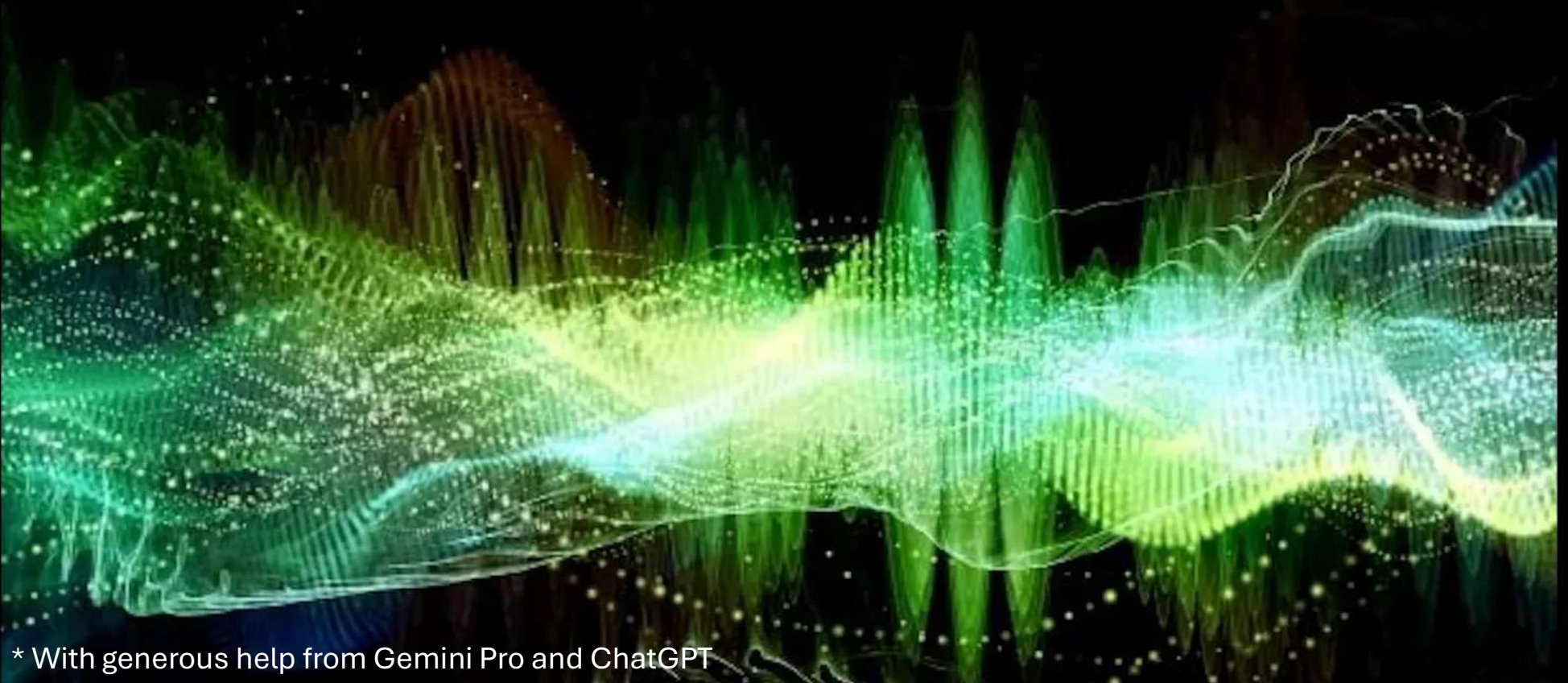


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 6, 9-10-26

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* With generous help from Gemini Pro and ChatGPT

Transfer Functions, Stability, and Pole–Zero Plots

Turn rational algebra into geometric intuition about filters.

poles push up
zeros pull down

75 minutes • rigorous DSP foundations • audio examples • Python sandbox • ML bridge

lec 5 $X(z)$ + ROC

lec 6 System analysis from rational $H(z)$

Learning outcomes

00-20 MIN

By the end, students should predict filter behavior before plotting.

- Derive $H(z)$ directly from a linear constant-coefficient difference equation.
- Identify poles, zeros, gain, and ROC constraints.
- Use z-domain rules to test causality and BIBO stability.
- Estimate frequency-response shape geometrically from a pole-zero plot.
- Design and test a hum notch and a resonator in Python.

demo

← estimate
 $H(e^{j\omega})$

Difference equation → transfer function

Zero initial conditions turn shifts into powers of z^{-1} .

Difference equation

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

degree $\max\{N, M\}$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

rational
system
function

Example

$N=2$ $M=1$ deg 2

$$\begin{aligned} y[n] - 1.2y[n-1] + 0.32y[n-2] \\ = x[n] + 0.3x[n-1] \end{aligned}$$

$$H(z) = (1 + 0.3z^{-1}) / (1 - 1.2z^{-1} + 0.32z^{-2})$$

- Numerator roots → zeros.
- Denominator roots → poles.
- Overall scale → gain.
- ROC is determined by the intended system support/causality, not merely by the polynomial ratio.

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

Delay property of DTFT

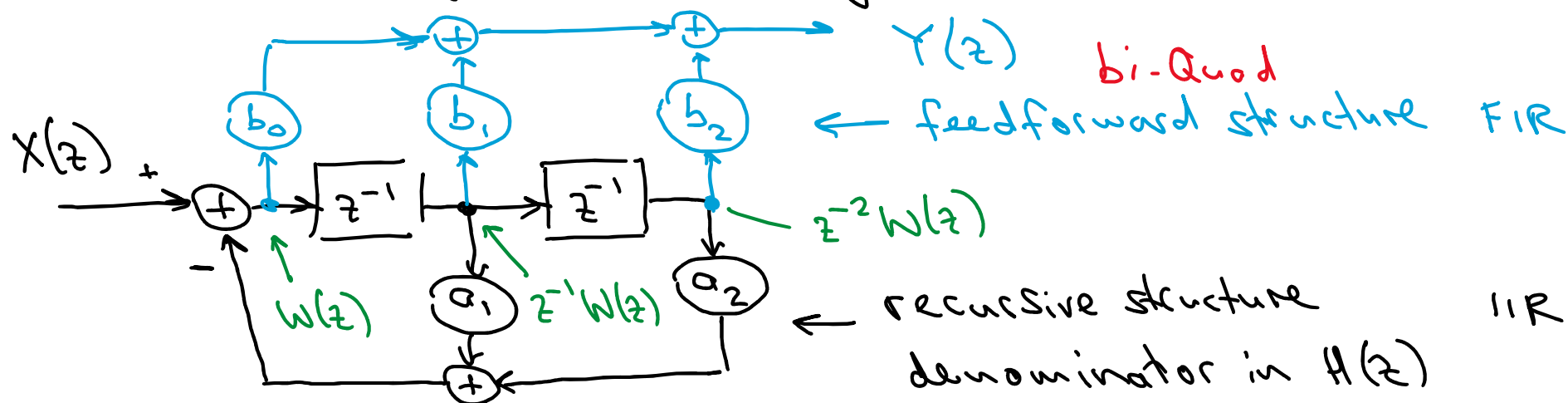
if $X(z) = \sum_n x[n] z^{-n}$ then

$$\sum_n x[n - n_0] z^{-n} = \sum_m x[m] z^{-m} z^{-n_0} = z^{-n_0} X(z)$$

$\begin{matrix} m & \uparrow & n - n_0 \\ n + n_0 & \uparrow & n \end{matrix}$

$$\left(\sum_{k=0}^N a_k z^{-k} \right) Y(z) = \left(\sum_{k=0}^M b_k z^{-k} \right) X(z) \Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}}$$

Full 2nd order system block diagram



$$W(z) = X(z) - a_1 z^{-1} W(z) - a_2 z^{-2} W(z)$$

$$\Rightarrow W(z)(1 + a_1 z^{-1} + a_2 z^{-2}) = X(z) \Rightarrow W(z) = \frac{1}{1 + a_1 z^{-1} + a_2 z^{-2}} X(z)$$

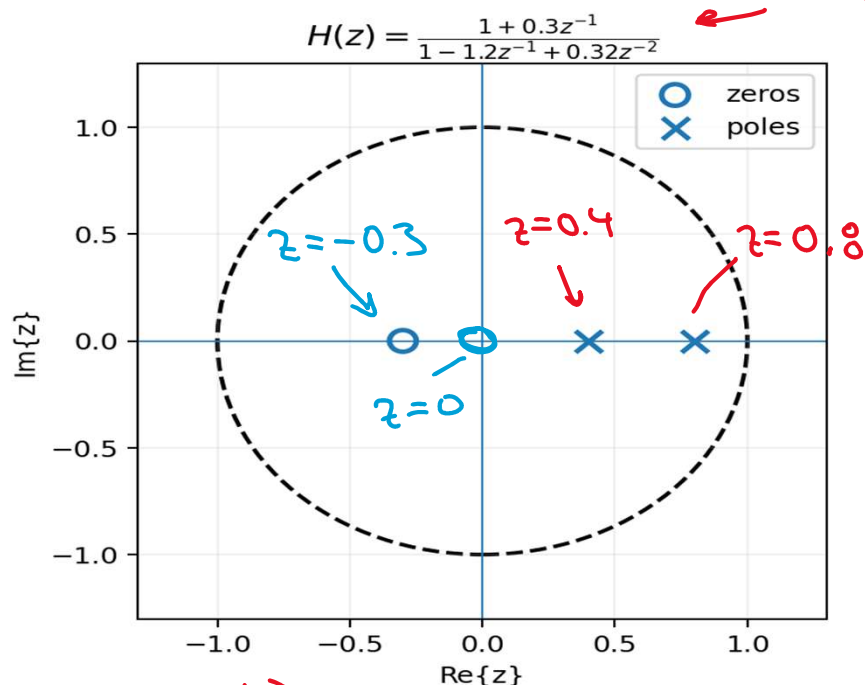
$$Y(z) = b_0 W(z) + b_1 z^{-1} W(z) + b_2 z^{-2} W(z) =$$

$$= \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}} X(z) \quad H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Factor H(z): poles and zeros are the important landmarks

00-20 MIN

Root locations are more informative than raw coefficients.



$$H(z) = \frac{1 + 0.3z^{-1}}{(1 - 0.4z^{-1})(1 - 0.8z^{-1})}$$

For the example

zero: $z = -0.3$

poles: $z = 0.8, 0.4$

All poles and zeros are distinct, so the pole-zero plot directly exposes the second-order dynamics.

Engineering habit: factor before you sketch.

multiply $H(z)$
by z^2

$$\Rightarrow H(z) = \frac{z \cdot (z + 0.3)}{(z - 0.4)(z - 0.8)}$$

zero at $z = 0$

System function $H(z)$
of degree N has
 N zeros and N poles

Causality in the z-domain

20-40 MIN

For a rational LTI system, a causal impulse response is right-sided.

Causal rule

ROC lies outside the outermost pole.

If poles are at radii 0.4 and 0.8:

causal ROC $\Rightarrow |z| > 0.8$

Why?

- Causal $h[n]$ is right-sided.
- Right-sided rational sequences have exterior ROCs.
- The outermost pole sets the convergence boundary.

$$H(z) = \frac{1 + 0.3z^{-1}}{(1 - 0.4z^{-1})(1 - 0.8z^{-1})} = \frac{A}{1 - 0.4z^{-1}} + \frac{B}{1 - 0.8z^{-1}}$$

$$h[n] = [A(0.4)^n + B(0.8)^n] u[n]$$

if ROC is $|z| > 0.8$

causal
(right sided)

takes longer to decay
dominant pole is
at $z = 0.8$

BIBO stability in the z-domain

20-40 MIN

For LTI systems, stability means the unit circle is in the ROC.

Time-domain condition

$$\sum |h[n]| < \infty$$

Then $H(e^{j\omega}) = \sum_n h[n]e^{-j\omega n}$ converges for every ω .

$$H(e^{j\omega}) = \sum_n h[n]e^{-j\omega n} \quad \text{DTFT}$$

exists iff $\sum_n |h[n]| < \infty$

in the
conventional
sense

z-plane condition

unit circle $\subset$ ROC

The frequency response is finite all the way around $|z|=1$.

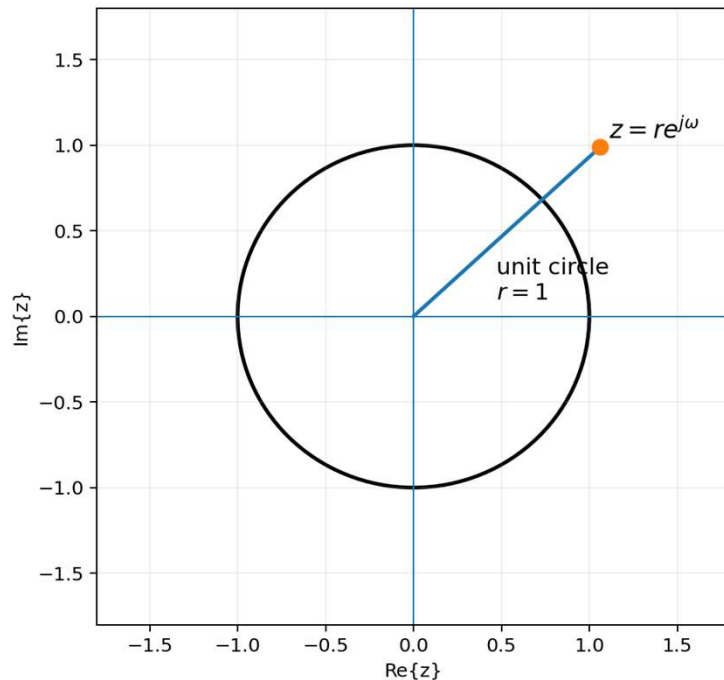
System is BIBO stable
iff unit circle is in ROC

The golden combination: causal + stable

20–40 MIN

For a causal rational system, every pole must lie strictly inside the unit circle.

For causal stable system all poles must lie inside unit circle



Reasoning chain

- Causal $\Rightarrow$ ROC is outside the outermost pole.
- Stable $\Rightarrow$ ROC must include $|z|=1$.
- Therefore outermost pole radius must be < 1 .
- Pole exactly on $|z|=1$ is not BIBO stable for the causal rational case.

Important nuance: “all poles inside unit circle” is the causal + stable rule—not a stability rule without an ROC/support assumption.

For ^{stable} anti-causal system All poles must lie outside unit circle

Classify three systems from poles + ROC

20-40 MIN

Require one sentence of justification.

A poles 0.3, 0.8; ROC $|z| > 0.8$

right sided causal
stable (unit circle in ROC)

B pole 1.1; ROC $|z| > 1.1$

causal
not stable

C pole 1.1; ROC $|z| < 1.1$

anti-causal
stable

Classify three systems from poles + ROC

20–40 MIN

Require one sentence of justification.

A poles 0.3, 0.8; ROC $|z| > 0.8$

Good answer: causal & stable

B pole 1.1; ROC $|z| > 1.1$

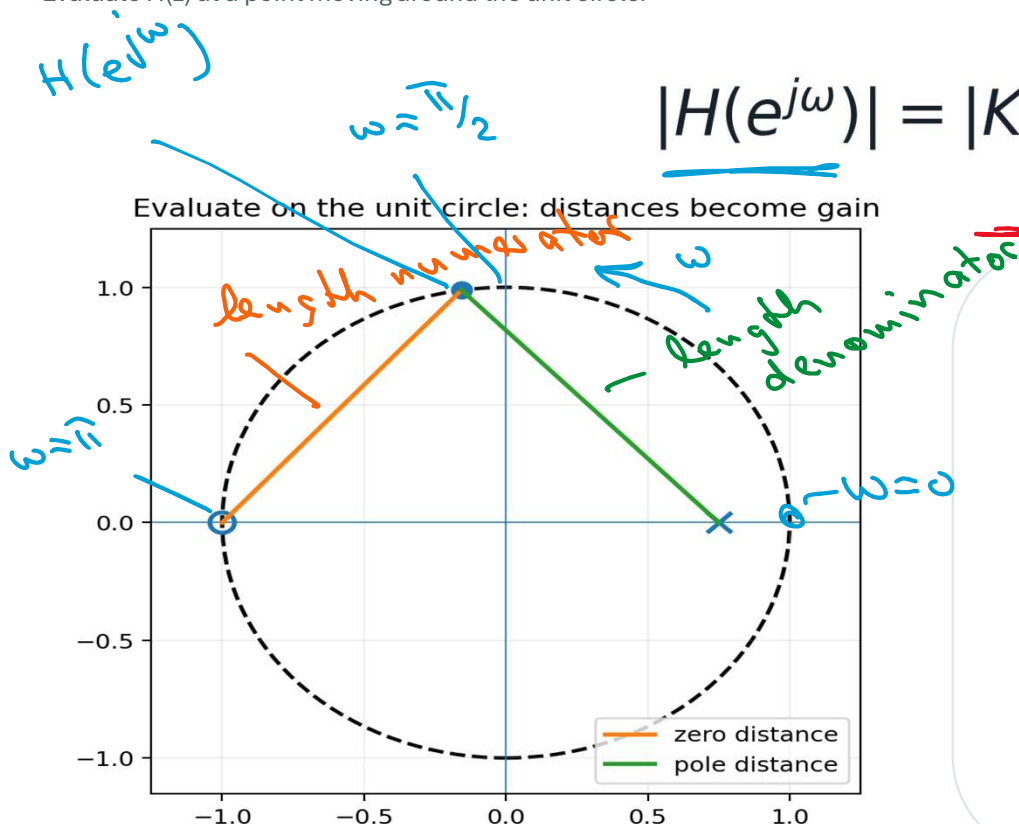
Good answer: causal, unstable

C pole 1.1; ROC $|z| < 1.1$

Good answer: anti-causal, stable

Geometric frequency response

Evaluate $H(z)$ at a point moving around the unit circle.



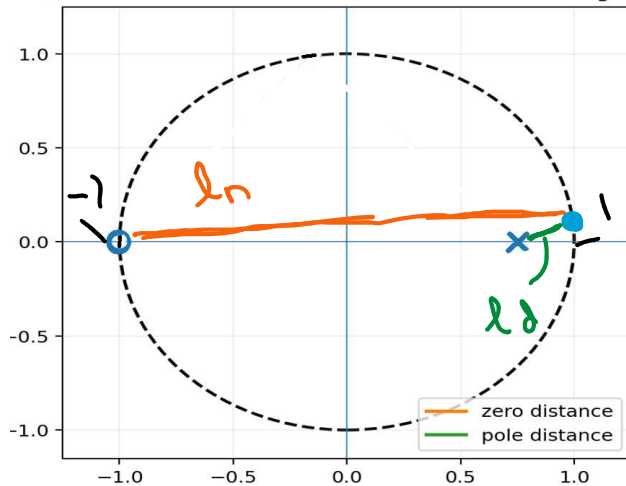
- Numerator: multiply distances from the unit-circle point to each zero.
- Denominator: multiply distances to each pole.
- Near a zero $\rightarrow$ a short numerator vector $\rightarrow$ small magnitude.
- Near a pole $\rightarrow$ a short denominator vector $\rightarrow$ large magnitude.

Geometric frequency response

$$|H(e^{j\omega})| = |K| \frac{\prod_k \overbrace{e^{j\omega} - z_k}^{\text{zero distance}}}{\prod_m \underbrace{e^{j\omega} - p_m}_{\text{pole distance}}}$$

small ω

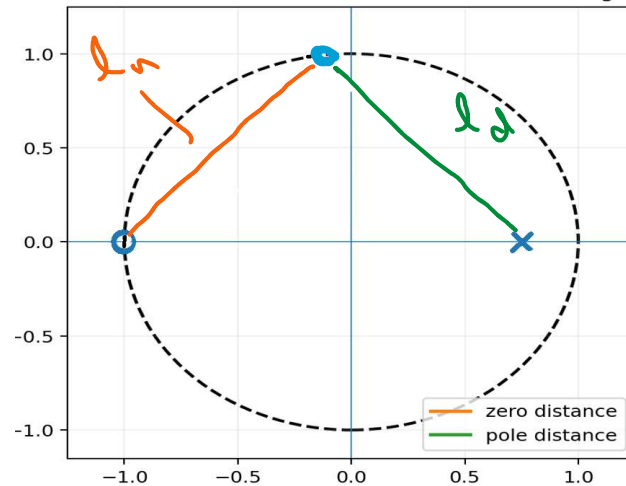
Evaluate on the unit circle: distances become gain



$|H(e^{j\omega})| = \text{large}$

medium ω

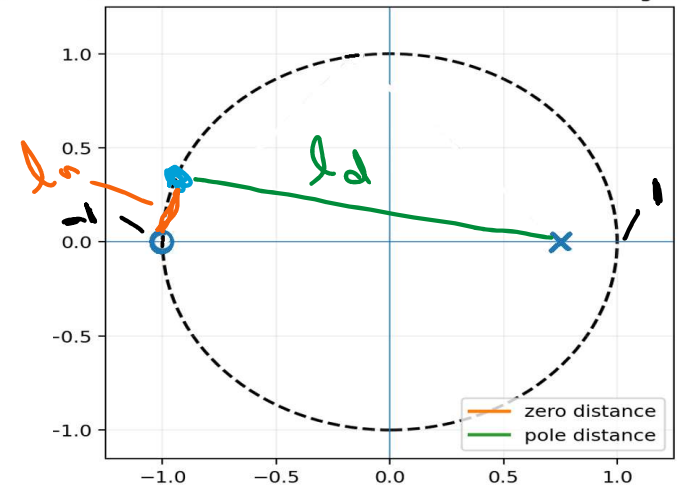
Evaluate on the unit circle: distances become gain



$|H(e^{j\omega})| \approx 1$

LPF

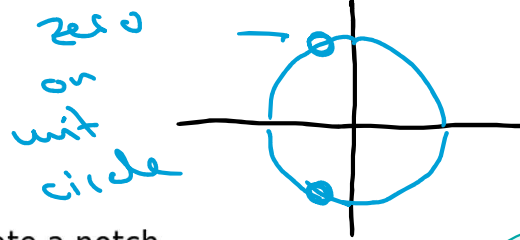
Evaluate on the unit circle: distances become gain



$|H(e^{j\omega})| \text{ small}$

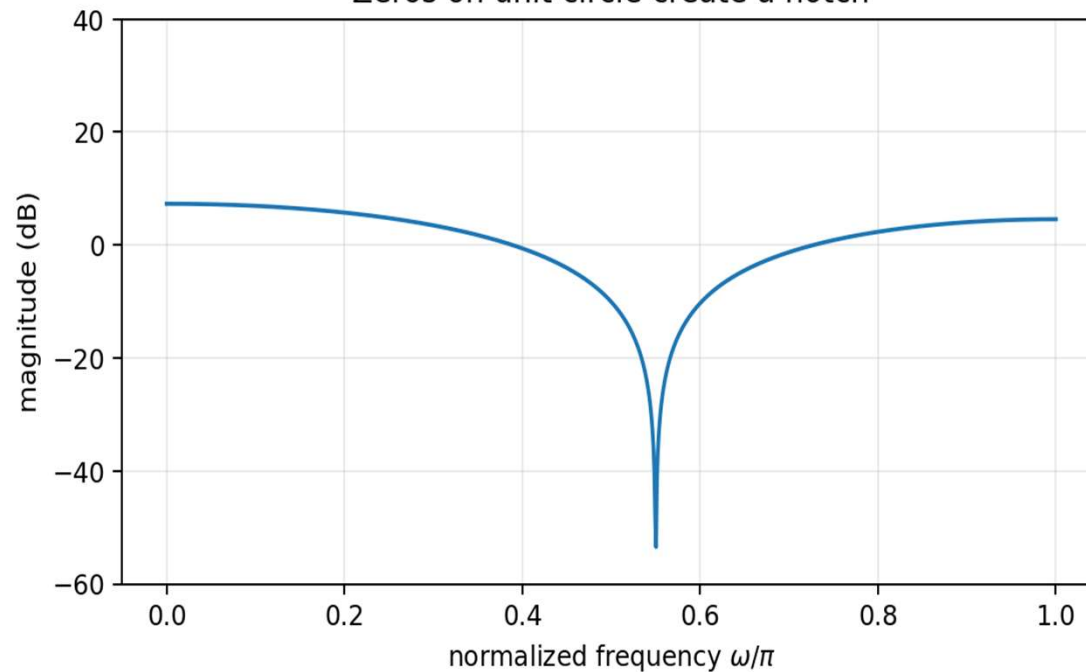
Zeros pull the magnitude down

A zero on the unit circle creates exact cancellation at its angle.



40–60 MIN

Zeros on unit circle create a notch

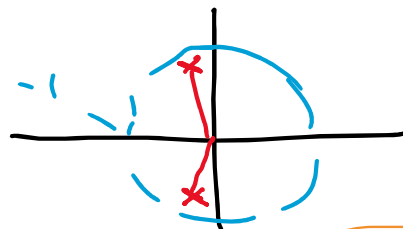


Geometric story

- At ω equal to the zero angle, $e^{j\omega} = z_0$
- The zero distance is exactly 0.
- Product in the numerator becomes 0.
- Therefore $|H|=0$: a perfect notch.

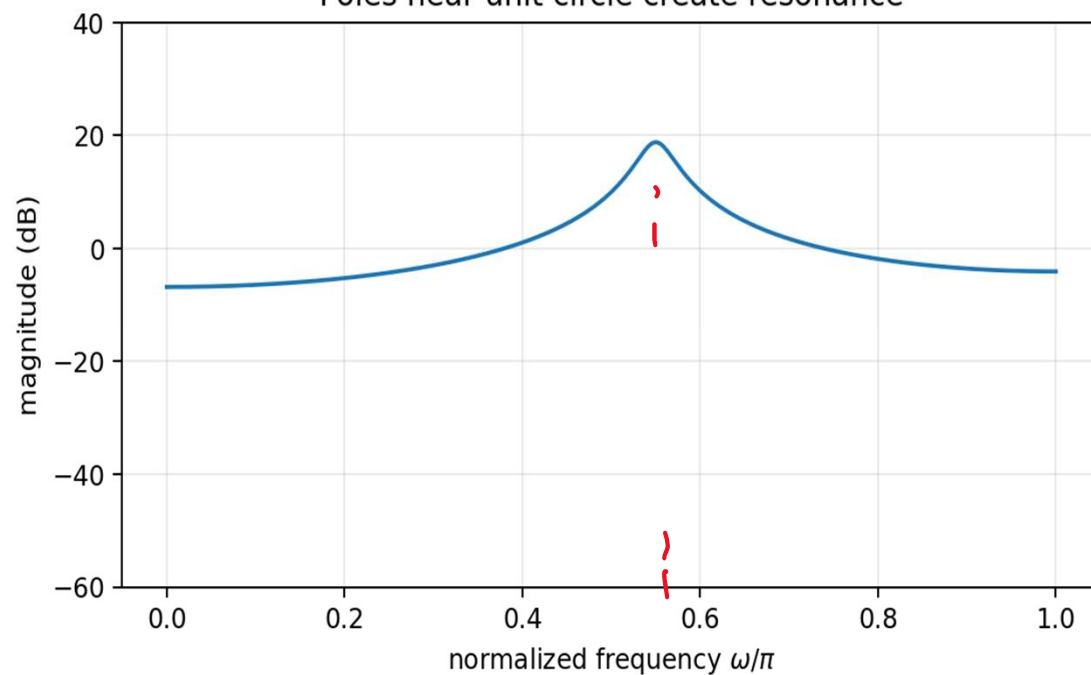
Poles push the magnitude up

A pole close to the unit circle creates a resonant peak near its angle.



40–60 MIN

Poles near unit circle create resonance



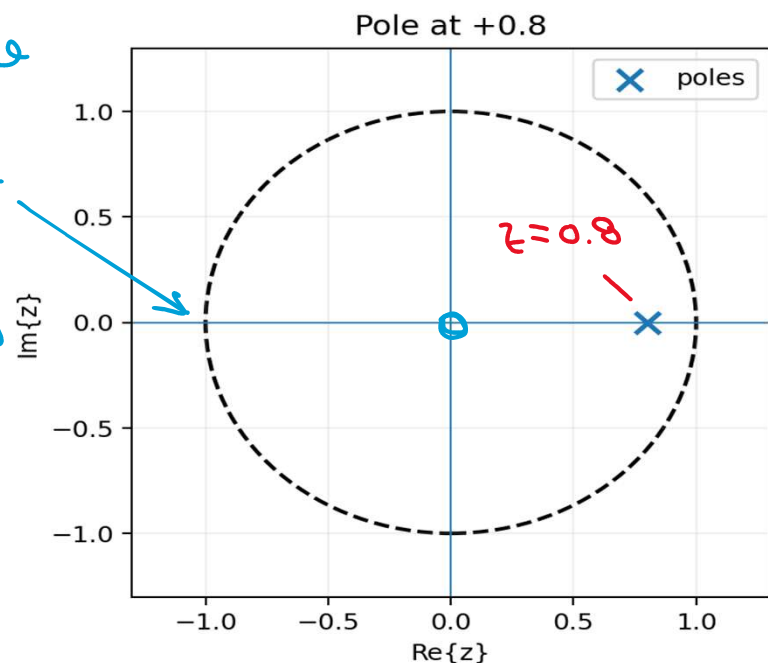
Geometric story

- At a nearby unit-circle angle, the pole distance becomes small.
- That distance is in the denominator.
- Smaller denominator $\rightarrow$ larger magnitude.
- Closer pole radius $\rightarrow$ sharper, longer-lived resonance.

First-order example: pole location predicts low-pass behavior

40-60 MIN

$H(z) = 1/(1 - 0.8z^{-1})$ has a pole at +0.8 on the real axis.



- $\omega=0$ corresponds to $z=+1$, close to the pole at +0.8 $\rightarrow$ large gain.
- $\omega=\pi$ corresponds to $z=-1$, far from the pole $\rightarrow$ smaller gain.
- So the response favors low frequency before we compute a single formula.
- This recovers the one-pole smoother from Week 1.

$$H(z) = \frac{1}{1 - 0.8z^{-1}}$$

$$\text{zero } z = 0$$

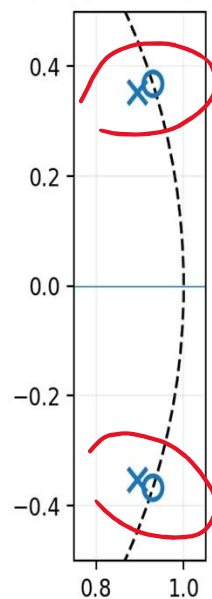
$$\text{pole } z = 0.8$$

Design example: notch out 60 Hz hum

60-75 MIN

Place conjugate zeros on the unit circle at the unwanted frequency

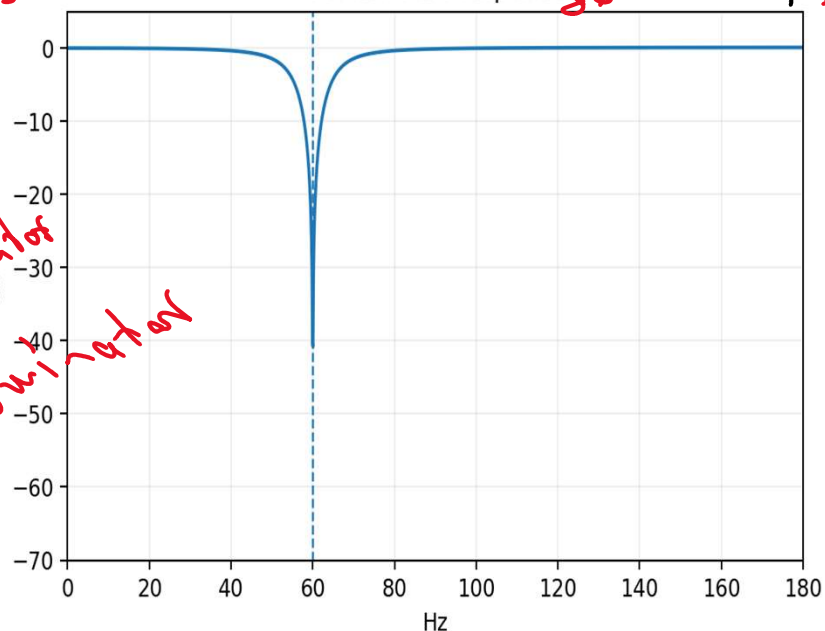
Notch: zeros at hum angle



*compl
conj
is
in
numerator
and
denominator*

*$\rho = 1$ for numerator
 $\rho < 1$ denominator*

60 Hz notch response



At $f_s = 1000$ Hz

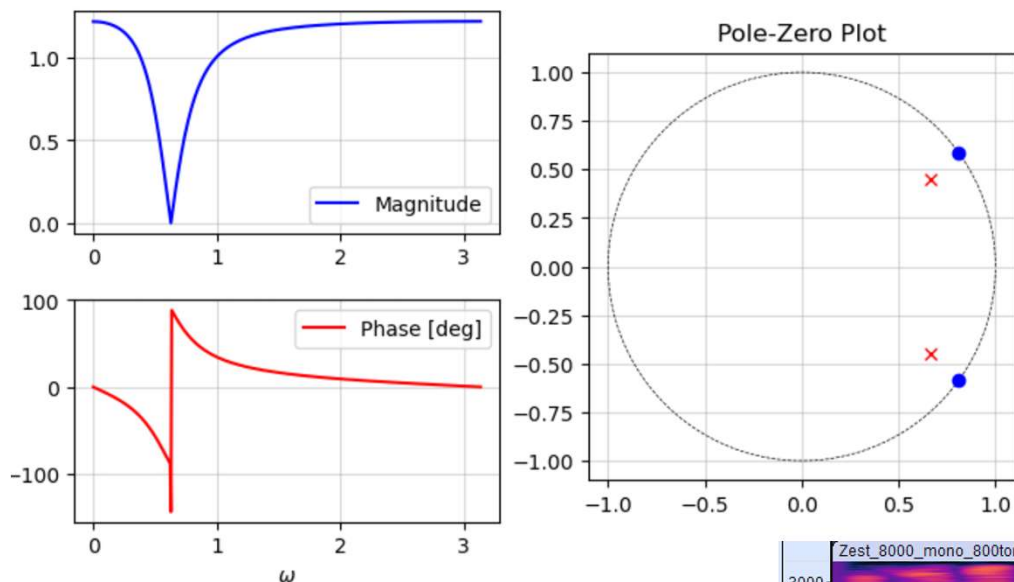
$$\omega_0 = 2\pi \cdot 60 / 1000 \approx 0.377 \text{ rad/sample}$$

- Zeros: $e^{\pm j\omega_0}$
- Nearby poles at $\rho e^{\pm j\omega_0}$ make a narrow, well-behaved notch.
- Choose $\rho < 1$ for causal stability.

complex conj pole/zero pair

$$P(z) = (1 - \rho e^{j\omega_0} z^{-1})(1 - \rho e^{-j\omega_0} z^{-1}) = 1 - 2\rho \cos(\omega_0) z^{-1} + \rho^2 z^{-2}$$

Design example: notch out 800 Hz tone



Notch filter:

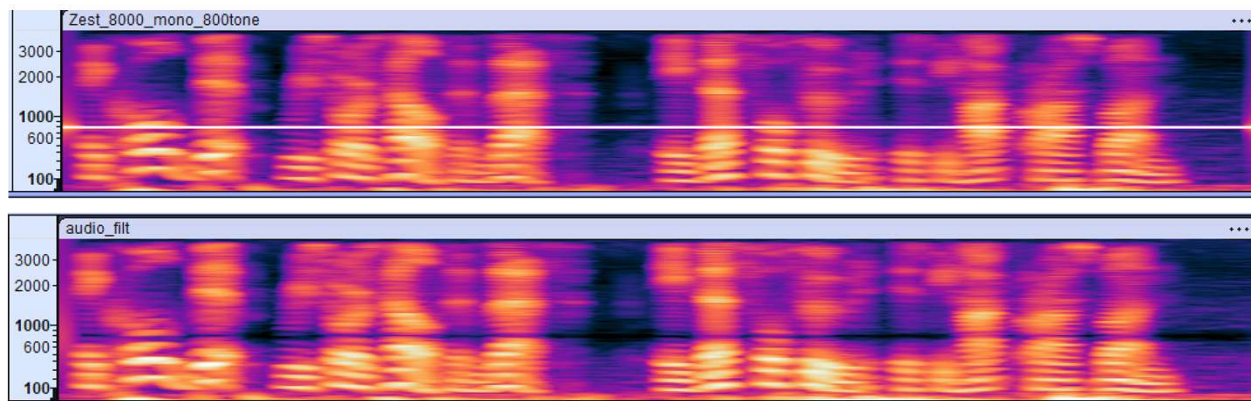
$$H(z) = \frac{1 - 2 \cos(\omega_z) z^{-1} + z^{-2}}{1 - 2r_p \cos(\omega_p) z^{-1} + r_p^2 z^{-2}}$$

$$\omega_z = 36^\circ, r_p = 0.8, \omega_p = 34^\circ, f_s = 8000 \text{ Hz}$$

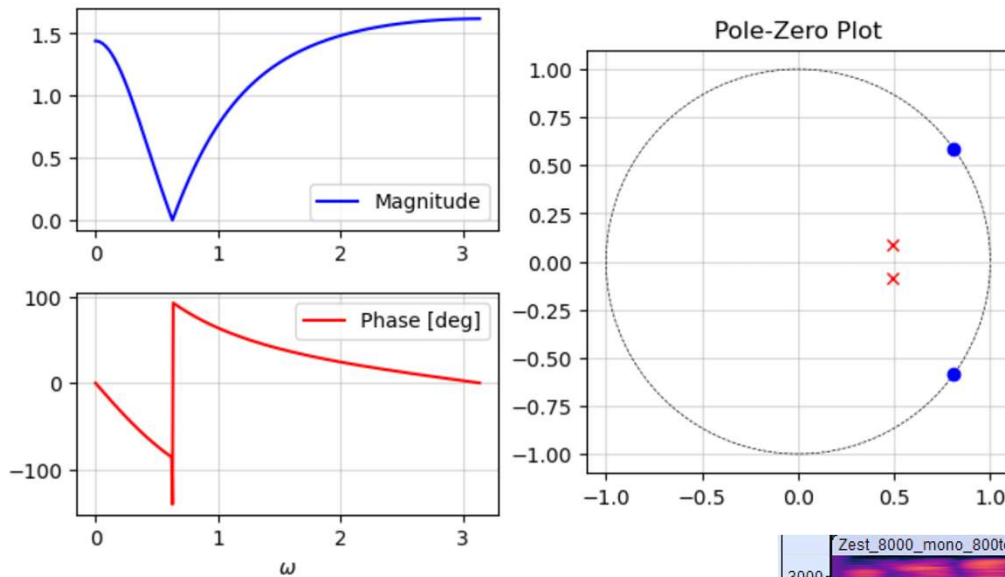
Before filtering

Spectrograms

After filtering



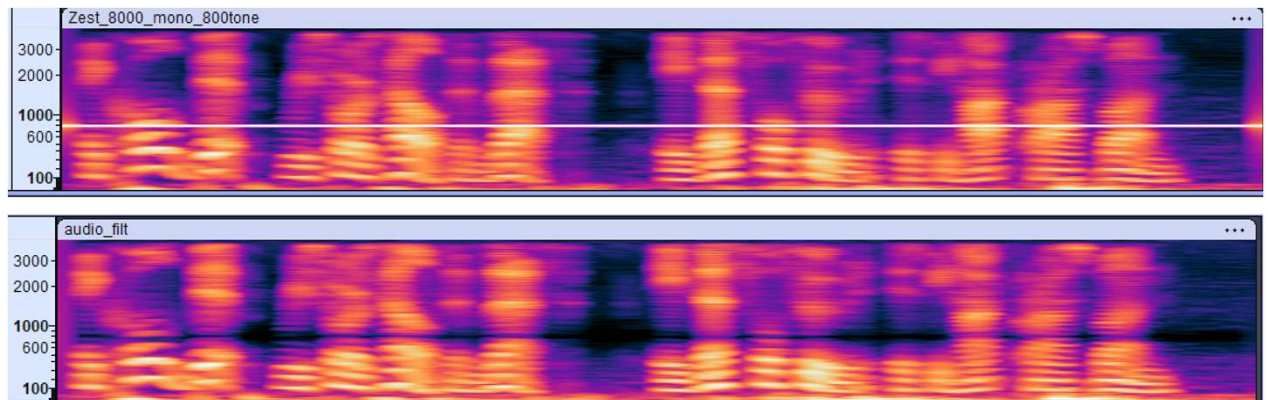
Design example: notch out 800 Hz tone



Before filtering

Spectrograms

After filtering



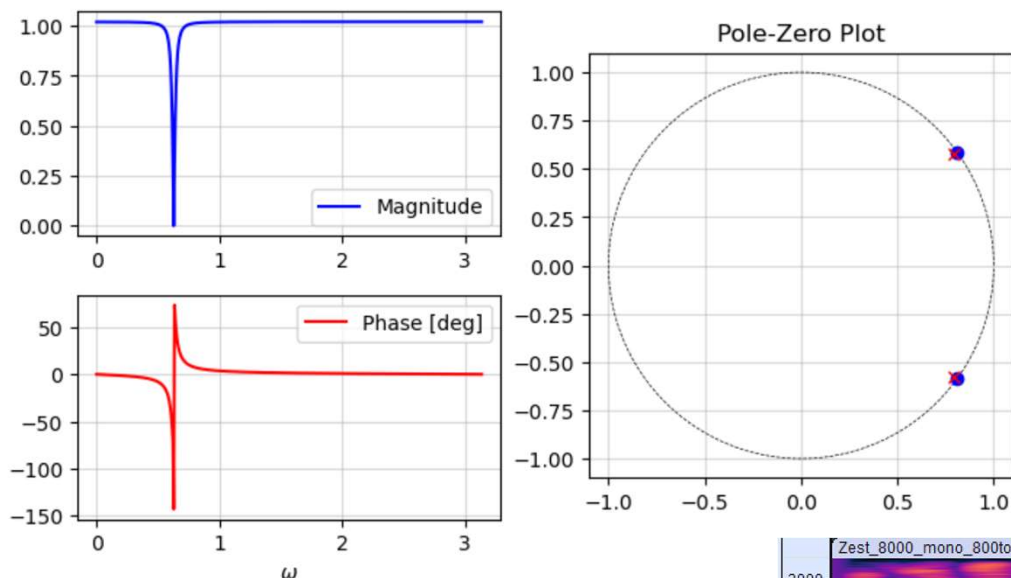
Notch filter:

$$H(z) = \frac{1 - 2 \cos(\omega_z) z^{-1} + z^{-2}}{1 - 2r_p \cos(\omega_p) z^{-1} + r_p^2 z^{-2}}$$

$$\omega_z = 36^\circ, r_p = 0.5, \omega_p = 10^\circ, f_s = 8000 \text{ Hz}$$

Smaller $r_p \Rightarrow$ Wider notch

Design example: notch out 800 Hz tone



Notch filter:

$$H(z) = \frac{1 - 2 \cos(\omega_z) z^{-1} + z^{-2}}{1 - 2r_p \cos(\omega_p) z^{-1} + r_p^2 z^{-2}}$$

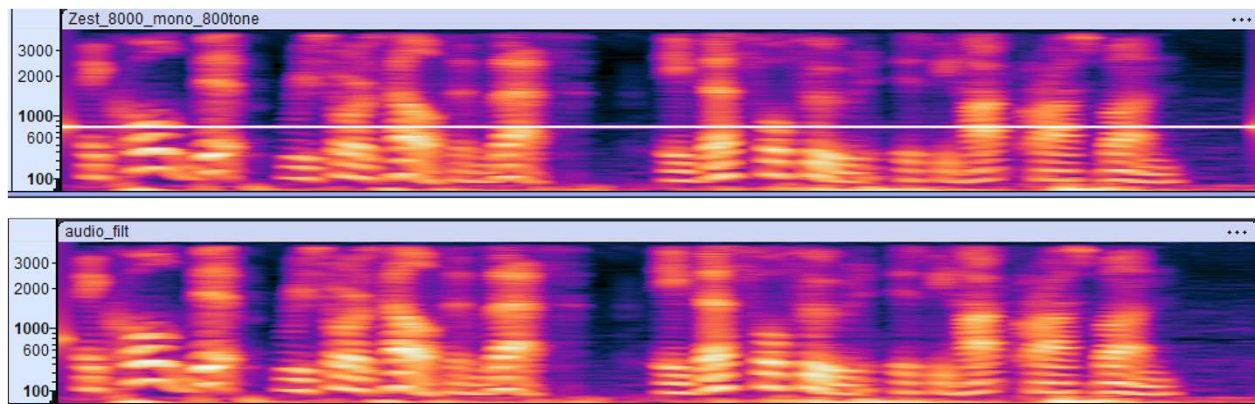
$$\omega_z = 36^\circ, r_p = 0.5, \omega_p = 10^\circ, f_s = 8000 \text{ Hz}$$

r_p close to unit circle $\Rightarrow$ Sharper notch

Before filtering

Spectrograms

After filtering



Notch-filter equation

60–75 MIN

A second-order section gives an interpretable audio filter.

$$H(z) = K \cdot (1 - 2\cos\omega_0 z^{-1} + z^{-2}) / (1 - 2r \cos\omega_0 z^{-1} + r^2 z^{-2})$$

Zeros

exactly on unit circle at $\pm\omega_0 \rightarrow$ ideal zero response at the hum frequency

Poles

same angle but radius $r < 1 \rightarrow$ control notch width / transient response

Gain K

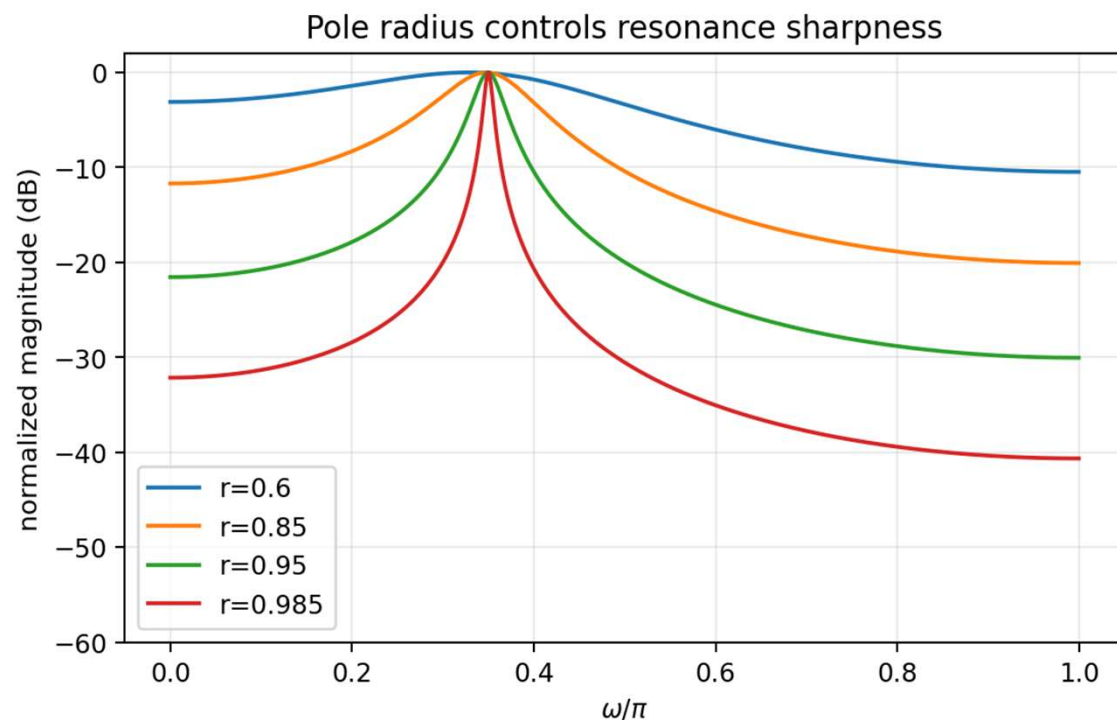
often chosen to make DC or passband gain approximately 1

Audio intuition: zeros do the cancellation; poles nearby keep the rest of the spectrum from being unnecessarily attenuated.

Pole radius controls resonance and memory

60–75 MIN

Move the pole radially without changing its preferred frequency.



- Same angle $\Rightarrow$ same resonance center.
- r closer to 1 $\Rightarrow$ smaller distance to unit circle.
- Peak becomes sharper and stronger.
- Impulse response decays roughly like r^n , so ringing lasts longer.
- $r \geq 1$ breaks causal BIBO stability.

Pole-zero sandbox activity

60–75 MIN

Predict first. Move a pole or zero. Then listen / plot.

1

**Start with no poles/zeros:
flat response.**

2

**Put a zero on $|z|=1$ at an
angle: predict the notch.**

3

**Move the zero inside the
circle: does the notch
remain perfect?**

4

**Add a pole at the same
angle with $r=0.9$, then 0.98 .**

5

**Listen to the 60-Hz hum
example before and after
filtering.**

Notebook: `pole_zero_sandbox()`

Python demo: poles, zeros, and freqz

60–75 MIN

Use code as a geometry checker.

```
import numpy as np
from scipy import signal

zeros = np.exp(1j*np.array([w0, -w0]))
poles = r * zeros
b = np.poly(zeros).real
a = np.poly(poles).real

w, H = signal.freqz(b, a, fs=fs)
# plot zeros, poles, |H|
```

- `np.roots` moves coefficients → roots.
- `np.poly` moves roots → coefficients.
- `signal.freqz` evaluates on the unit circle numerically.
- The pole-zero plot should let you predict the major features before `freqz` confirms them.

Audio demo: removing hum without erasing the signal

60–75 MIN

The notch should target a narrow frequency, not broadly low-pass the recording.

Before

Synthetic voice-like harmonics + 60 Hz and 120 Hz hum.
Listen: 01_voice_with_60Hz_hum.wav

After

Second-order 60-Hz notch with pole radius $r=0.985$.
Listen: 02_voice_after_60Hz_notch.wav

Good observation: a very narrow notch removes the targeted hum component while leaving nearby useful content mostly intact. A broad low-pass filter would solve the hum problem by throwing away far more information.

ML bridge: filter geometry shapes the features a model sees

BRIDGE

Front-end filtering changes spectra, spectrograms, and therefore learned patterns.

Notch zeros

Remove a nuisance line such as power hum before spectral feature extraction.

Resonant poles

Can emphasize a narrow band—useful or dangerous depending on the task.

Stability

A numerically exploding front end corrupts every downstream feature and gradient.

Design question for ML: are you removing a known nuisance, imposing a useful inductive bias, or accidentally deleting information that the classifier needs?

Concept check

CHECK

Reason geometrically—no full algebra is needed.

A conjugate pole pair is moved from radius 0.7 to radius 0.97 at the same angle. What changes?

A Resonant frequency moves drastically.

B Peak becomes sharper; ringing lasts longer.

C A perfect notch appears.

D The system becomes time varying.

Concept check

CHECK

Reason geometrically—no full algebra is needed.

A conjugate pole pair is moved from radius 0.7 to radius 0.97 at the same angle. What changes?

A Resonant frequency moves drastically.

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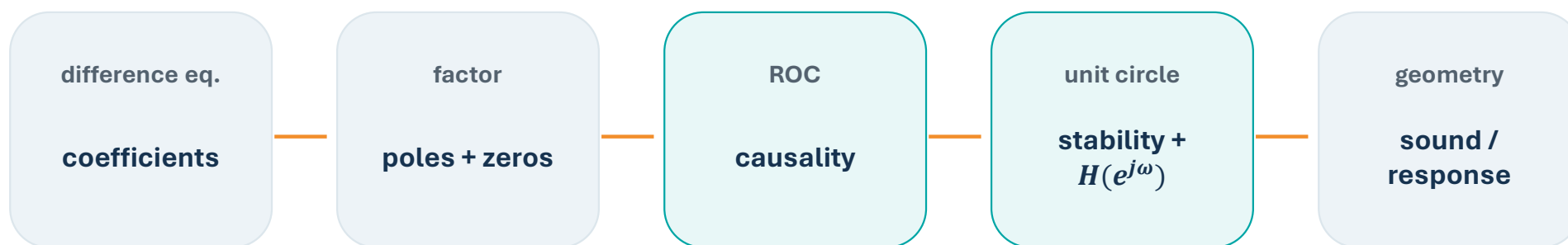
D The system becomes time varying.

Correct: B. Same angle $\rightarrow$ same center frequency; radius closer to 1 $\rightarrow$ stronger/narrower resonance and slower decay.

Lecture 6 synthesis

WRAP-UP

Poles, zeros, ROC, and the unit circle tell one coherent story.

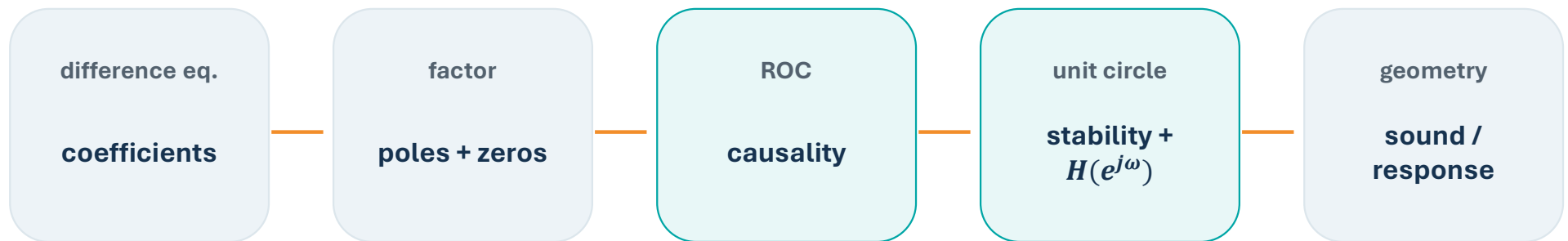


Exit ticket: Why does a zero on the unit circle create a notch?

Lecture 6 synthesis

WRAP-UP

Poles, zeros, ROC, and the unit circle tell one coherent story.



Exit ticket: Why does a zero on the unit circle create a notch? Good answer: at that frequency the evaluation point $e^{j\omega}$ coincides with the zero, making a numerator distance—and therefore $H(e^{j\omega})$ —equal to zero.